INJECTIVITY OF THE OMEGA-FIXED CENTRAL-EXTENSION PRESENTATION INSIDE THE UNIVERSAL CENTRAL EXTENSION OF SL(N)[U^{+-1},V]

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Allowed standard prerequisites (editorial).

- PBW theorem, affine root systems, universal central extensions. Original MRY, GNW and HU bibliography entries; author formulations and uses preserved

1. Universal central extension of $\mathfrak{sl}(n)\big[u^{\pm 1}, v\big]$

In this section, we recall the work of [18], which gives a presentation of the universal central extension of $\mathfrak{sl}(n)\big[u^{\pm 1}, v\big]$. For a perfect Lie algebra $\mathfrak{g}$, a central extension of $\mathfrak{g}$ is a pair consisting of a Lie algebra $\widetilde{\mathfrak{g}}$ and a surjective homomorphism $\pi\colon \widetilde{\mathfrak{g}} \to \mathfrak{g}$ whose kernel is contained in the center of $\widetilde{\mathfrak{g}}$. The central extension $(\widetilde{\mathfrak{g}}, \pi)$ is called the universal central extension of $\mathfrak{g}$ if for every central extension $(\mathfrak{e}, \psi)$ there exists a unique homomorphism $\phi\colon \widetilde{\mathfrak{g}} \to \mathfrak{e}$ satisfying that $\psi \circ \phi = \pi$.

In [11], Kassel gave an explicit presentation of the universal central extension of $\mathfrak{sl}(n)\big[u^{\pm 1}, v\big]$. We set $\big(\Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big), d\big)$ as the module of differentials of $\mathbb{C}\big[u^{\pm 1}, v\big]$ in the same way as in [18, Section 2]. We take $\{a_i\}$ as a basis of $\mathbb{C}\big[u^{\pm 1}, v\big]$. Let F be a free module of $\mathbb{C}\big[u^{\pm 1}, v\big]$ whose basis is $\big\{\widetilde{d}a_i\big\}$. We regard F as a 2-sided $\mathbb{C}\big[u^{\pm 1}, v\big]$ module by setting $a\big(\widetilde{d}b\big) = \big(\widetilde{d}b\big)a$ for any $a, b \in \mathbb{C}\big[u^{\pm 1}, v\big]$. We set a linear map $\widetilde{d}\colon \mathbb{C}\big[u^{\pm 1}, v\big] \to F$ as $\widetilde{d}\big(\sum_i c_i a_i\big) = \sum_i c_i \widetilde{d}a_i$ and K as a submodule of F generated by $\widetilde{d}(ab) = (\widetilde{d}a)b + a\big(\widetilde{d}b\big)$ for $a, b \in \mathbb{C}\big[u^{\pm 1}, v\big]$. We denote F/K by $\Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)$. We also set $d\colon \mathbb{C}\big[u^{\pm 1}, v\big] \to \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)$ as a homomorphism induced by $\widetilde{d}$.

Let $\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big] \oplus \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]$ be the Lie algebra whose commutator relations are given by

$$
\begin{aligned}
[z_1 \otimes a_1, z_2 \otimes a_2] &= [z_1, z_2] \otimes a_1 \cdot a_2 + \kappa(z_1, z_2) a_2 d a_1, \\
\big[\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big], \Omega^1\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]\big] &= 0, \\
\big[\Omega^1\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big], \Omega^1\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]\big] &= 0.
\end{aligned}
$$

We can naturally define a surjective homomorphism

$$
\pi\colon\ \mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big] \oplus \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big] \to \mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big].
$$

Since the kernel of π is $\Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]$, π gives a central extension. Moreover, by [11], $\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big] \oplus \Omega\big(\mathbb{C}\big[u^{\pm 1}, v\big]\big)/d\mathbb{C}\big[u^{\pm 1}, v\big]$ together with π form the universal central extension of $\mathfrak{sl}(n) \otimes \mathbb{C}\big[u^{\pm 1}, v\big]$.

Moody–Rao–Yokonuma [18] gave a new presentation of this universal central extension, which we describe below. Let us take a matrix $\big(a^A_{i,j}\big)_{0 \le i,j \le n-1}$ as

$$
a^A_{i,j} = \begin{cases} 2 & \text{if } i = j, \\ -1 & \text{if } i = j \pm 1, \\ -1 & \text{if } (i,j) = (0, n-1), (n-1, 0), \\ 0 & \text{otherwise.} \end{cases}
$$

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Then, $\bigl(a^A_{i,j}\bigr)_{0\le i,j\le n-1}$ is the Cartan matrix of an affine Lie algebra $\widehat{\mathfrak{sl}}(n) = \mathfrak{sl}(n) \otimes \mathbb{C}\bigl[t^{\pm 1}\bigr] \oplus \mathbb{C}c$ whose commutator relations are given by

$$[x \otimes t^u, y \otimes t^s] = [x, y] \otimes t^{u+s} + u\delta_{u+s,0} \operatorname{tr}(xy)c \qquad \text{for } x, y \in \mathfrak{sl}(n),$$
$$\bigl[c, \widehat{\mathfrak{sl}}(n)\bigr] = 0,$$

where tr is the trace of $\mathfrak{gl}(n)$.

Definition 1.1. We define $\mathfrak{A}(n)$ as the Lie algebra over $\mathbb{C}$ generated by

$$\bigl\{x^{\pm}_{i,r}, h_{i,r} \mid 0 \le i \le n-1,\ r \in \mathbb{Z}_{\ge 0}\bigr\}$$

subject to the following defining relations:

$$[h_{i,r}, h_{j,s}] = 0, \tag{1.1}$$
$$\bigl[x^{+}_{i,r}, x^{-}_{j,s}\bigr] = \delta_{i,j} h_{i,r+s}, \tag{1.2}$$
$$\bigl[h_{i,0}, x^{\pm}_{j,r}\bigr] = \pm a^A_{i,j} x^{\pm}_{j,r}, \tag{1.3}$$
$$\bigl[h_{i,r+1}, x^{\pm}_{j,s}\bigr] = \bigl[h_{i,r}, x^{\pm}_{j,s+1}\bigr], \tag{1.4}$$
$$\bigl[x^{\pm}_{i,r+1}, x^{\pm}_{j,s}\bigr] = \bigl[x^{\pm}_{i,r}, x^{\pm}_{j,s+1}\bigr], \tag{1.5}$$
$$\sum_{\sigma \in S_{1-a_{i,j}}} \bigl(\operatorname{ad} x^{\pm}_{i,r_{\sigma(1)}}\bigr) \cdots \bigl(\operatorname{ad} x^{\pm}_{i,r_{\sigma(1-a_{i,j})}}\bigr)\bigl(x^{\pm}_{j,s}\bigr) = 0 \qquad \text{if } i \ne j, \tag{1.6}$$

where S_l is a symmetric group of degree l. We also denote the universal enveloping algebra of $\mathfrak{A}(n)$ by $A(n)$.

By using (1.5), we obtain

$$\bigl[x^{\pm}_{i,r}, x^{\pm}_{i,s}\bigr] = \frac{1}{2}\bigl(\bigl[x^{\pm}_{i,r+s}, x^{\pm}_{i,0}\bigr] + \bigl[x^{\pm}_{i,0}, x^{\pm}_{i,r+s}\bigr]\bigr) = 0.$$

Moody–Rao–Yokonuma showed that $\mathfrak{A}(n)$ is isomorphic to the universal central extension of $\mathfrak{sl}(n) \otimes \mathbb{C}\bigl[u^{\pm 1}, v\bigr]$.

Theorem 1.2 ([18, Proposition 3.5])**.** *There exists an isomorphism*

$$\iota\colon\ \mathfrak{A}(n) \to \mathfrak{sl}(n) \otimes \mathbb{C}\bigl[u^{\pm 1}, v\bigr] \oplus \Omega\bigl(\mathbb{C}\bigl[u^{\pm 1}, v\bigr]\bigr)/d\mathbb{C}\bigl[u^{\pm 1}, v\bigr]$$

given by

$$x^{+}_{i,r} = \begin{cases} E_{i,i+1}v^r & \text{if } i \ne 0, \\ E_{n,1}uv^r & \text{if } i = 0, \end{cases} \qquad x^{-}_{i,r} = \begin{cases} E_{i+1,i}v^r & \text{if } i \ne 0, \\ E_{1,n}u^{-1}v^r & \text{if } i = 0, \end{cases}$$

where $E_{i,j}$ is a matrix unit of $\mathfrak{gl}(n)$ whose (p,q) component is $\delta_{p,i}\delta_{q,j}$.

Finally, we consider a presentation of the subalgebra of $A(n)$ generated by $\bigl\{x^{+}_{i,r}\bigr\}$. We take a subspace $\Omega_+ \subset \Omega\bigl(\mathbb{C}\bigl[u^{\pm 1}, v\bigr]\bigr)/d\mathbb{C}\bigl[u^{\pm 1}, v\bigr]$ spanned by $\bigl\{u^{r-1}v^s du \mid r \ge 1,\ s \ge 0\bigr\}$. We set a subalgebra $A(n)_+ \subset A(n)$ generated by $\bigoplus_{s\ge 0}\bigoplus_{\alpha\in\Delta^+} \mathfrak{g}_\alpha v^s$ and Ω_+. Let us take a grading on $A(n)$ by $\deg(h_{i,r}) = \deg\bigl(x^{\pm}_{i,r}\bigr) = r+1$ and denote the graded algebra associated with $A(n)$ by $\operatorname{gr} A(n)$.

Theorem 1.3. *The associative algebra $A(n)_+$ is isomorphic to the associative algebra generated by $\bigl\{\bar{x}^{+}_{i,r} \mid 1 \le i \le n-1,\ r \ge 0\bigr\}$ with the relations:*

$$\bigl[\bar{x}^{+}_{i,r+1}, \bar{x}^{+}_{j,s}\bigr] = \bigl[\bar{x}^{+}_{i,r}, \bar{x}^{+}_{j,s+1}\bigr],$$
$$\bigl[\bar{x}^{+}_{i,r}, \bar{x}^{+}_{j,s}\bigr] = 0 \qquad \text{if } a^A_{i,j} = 0,$$
$$\bigl[\bar{x}^{+}_{i,r}, \bigl[\bar{x}^{+}_{i,s}, \bar{x}^{+}_{j,u}\bigr]\bigr] = 0 \qquad \text{if } a^A_{i,j} = -1.$$

Let $\mathfrak{A}(n)_{\mathrm{pre}}$ be the Lie algebra generated by $\{h_{i,r}, x^{\pm}_{i,r} \mid 1 \le i \le n-1,\, r \ge 0\}$ with the defining relations (1.1)–(1.5). We also define $\mathfrak{A}(n)_{\mathrm{pre},\pm}$ and $\mathfrak{A}(n)_{\mathrm{pre},0}$ as Lie subalgebras generated by $\{x^{\pm}_{i,r}, h_{i,r} \mid 1 \le i \le n-1,\, r \ge 0\}$. Theorem 1.3 follows from the following lemma.

Lemma 1.4.

(1) *The associative algebra $U(\mathfrak{A}(n)_{\mathrm{pre},\pm})$ is isomorphic to a quotient algebra of the free associative algebra generated by $\{x^{\pm}_{i,r}\}$ divided by* (1.5).
(2) *Let $I_{\pm}$ be an ideal of $\mathfrak{A}(n)_{\mathrm{pre}}$ generated by* (1.6). *Then, $I_- \cap A(n)_{\mathrm{pre},+} = \varnothing$.*

Proof. (1) Let V be a quotient algebra of a noncommutative polynomial ring $\mathbb{C}\langle y_{i,d} \mid 0 \le i \le n-1, d \in \mathbb{Z}_{\ge 0}\rangle$ divided by the relations $[y_{i,r+1}, y_{j,s}] - [y_{i,r}, y_{j,s+1}]$. By a direct computation, we obtain a homomorphism

$$\kappa\colon\ \mathfrak{A}(n)_{\mathrm{pre}} \to \mathfrak{gl}(V)$$

by

$$\big(\kappa\big(x^{-}_{i,d}\big)\big)(y_{i_1,d_1}\cdots y_{i_r,d_r}) = y_{i,d}y_{i_1,d_1}\cdots y_{i_r,d_r},$$

$$(\kappa(h_{i,d}))(y_{i_1,d_1}\cdots y_{i_r,d_r}) = -\sum_{u=1}^{r} a^{A}_{i,i_u} y_{i_1,d_1}\cdots y_{i_{u-1},d_{u-1}} y_{i_u,d+d_u} y_{i_{u+1},d_{u+1}}\cdots y_{i_r,d_r},$$

$$\big(\kappa\big(x^{+}_{i,d}\big)\big)(1) = 0,$$

$$\big(\kappa\big(x^{+}_{i,d}\big)\big)(y_{i_1,d_1}\cdots y_{i_r,d_r}) = \delta_{i,i_1}\kappa(h_{i,d+d_1})(y_{i_2,d_2}\cdots y_{i_r,d_r}) + y_{i_1,d_1}\big(\kappa\big(x^{+}_{i,d}\big)\big)(y_{i_2,d_2}\cdots y_{i_r,d_r}).$$

The homomorphism κ induces a surjective homomorphism $\Xi\colon \mathfrak{A}(n)_{\mathrm{pre},-} \to V$ by $\Xi(x) = (\kappa(x))1$. By a direct computation, we can also construct a homomorphism

$$\xi\colon\ V \to \mathfrak{A}(n)_{\mathrm{pre},-}, \qquad y_{i,d} \mapsto x^{-}_{i,d}.$$

Since $\xi \circ \Xi$ is an identity map, Ξ is bijective. Then, we find that $U(\mathfrak{A}(n)_{\mathrm{pre},-})$ is isomorphic to a quotient algebra of the free associative algebra generated by $\{x^{-}_{i,r}\}$ divided by (1.5). Since $\mathfrak{A}(n)_{\mathrm{pre},+}$ is isomorphic to $\mathfrak{A}(n)_{\mathrm{pre},-}$, $U(\mathfrak{A}(n)_{\mathrm{pre},+})$ is also isomorphic to a quotient algebra of the free associative algebra generated by $\{x^{+}_{i,r} \mid r \ge 0\}$ divided by (1.5).

(2) It is enough to show that I_- is contained in $\mathfrak{A}(n)_{\mathrm{pre},-}$. This follows from the fact that $\mathrm{ad}\big(x^{+}_{i,r}\big)$ acts on the left-hand side of (1.6) trivially. □

2. A central extension of $\mathfrak{sl}(n)\big[u^{\pm 1}, v\big]^{\omega}$

First, we set notations about the affine Lie algebra $\widehat{\mathfrak{sl}}(n)$. Let Δ (resp. Δ^+, Δ^-) be the set of roots (resp. positive roots, negative roots) of $\widehat{\mathfrak{sl}}(n) = \mathfrak{sl}(n)\big[u^{\pm 1}\big] \oplus \mathbb{C}c$. We set an inner product $(\ ,\ )$ on $\widehat{\mathfrak{sl}}(n)$ by

$$(xu^s, yu^r) = \mathrm{tr}(xy)\delta_{r+s,0}, \qquad (xu^r, c) = (c,c) = 0.$$

We denote by $\mathfrak{g}_\alpha$ the root space associated with a root α. For $\alpha \in \Delta$, we take $\{x^k_\alpha\}_{k=1}^{\dim(\mathfrak{g}_\alpha)}$ as a basis of $\mathfrak{g}_\alpha$ such that $\big(x^k_\alpha, x^l_\beta\big) = \delta_{\alpha+\beta,0}\delta_{k,l}$ and $\big(x^k_\alpha\big)^{\mathsf{T}} = x^k_{-\alpha}$, where X^{T} is the transpose of a matrix X. The following relations hold in $U\big(\widehat{\mathfrak{sl}}(n)\big)^{\otimes 2}$ (see [10, Corollary 2.4]):

$$\sum_{k=1}^{\dim \mathfrak{g}_\alpha} \big[x^k_{-\alpha}, x_\beta\big] \otimes x^k_\alpha = \sum_{k=1}^{\dim \mathfrak{g}_{\beta-\alpha}} x^k_{\beta-\alpha} \otimes \big[x_\beta, x^k_{\alpha-\beta}\big]. \tag{2.1}$$

Let $\widehat{\tau}$ be an automorphism of $U\big(\widehat{\mathfrak{sl}}(n)\big)$ defined by $X \mapsto -X^{\mathsf{T}}$. Considering the images of both sides of (2.1) via $\widehat{\tau} \otimes \mathrm{id}$, we obtain

$$\sum_{k=1}^{\dim \mathfrak{g}_\alpha} \big[x^k_\alpha, x_{-\beta}\big] \otimes x^k_\alpha = -\sum_{k=1}^{\dim \mathfrak{g}_{\beta-\alpha}} x^k_{\alpha-\beta} \otimes \big[x_\beta, x^k_{\alpha-\beta}\big]. \tag{2.2}$$

Let us take an involution of $\mathfrak{sl}(n)\big[u^{\pm 1},v\big]$:

$$\omega\colon\ \mathfrak{sl}(n)\big[u^{\pm 1},v\big]\to\mathfrak{sl}(n)\big[u^{\pm 1},v\big],\qquad Xu^rv^s\mapsto -(-1)^sX^{\mathsf{T}}u^{-r}v^s.$$

In this section, we give a central extension of $\mathfrak{sl}(n)\big[u^{\pm 1},v\big]^{\omega}$.

Definition 2.1. Let $\mathfrak{BD}(n)$ be the Lie algebra generated by

$$\{H_{i,2r+1},B_{i,r}\mid 0\le i\le n-1,\ r\ge 0\}$$

with the following relations:

$$\begin{aligned}
[H_{i,2r+1},H_{j,2s+1}]&=0,\\
[H_{i,2r+1},B_{j,s}]&=2a^A_{i,j}B_{j,2r+s+1},\\
[B_{i,r+1},B_{j,s}]-[B_{i,r},B_{j,s+1}]&=-\delta_{i,j}((-1)^r+(-1)^s)H_{i,r+s},\\
[B_{i,r},B_{j,s}]&=0\qquad\text{if } a^A_{i,j}=0,\\
[B_{i,r},[B_{i,s},B_{j,u}]]&=-B_{j,r+s+u}\qquad\text{if } a^A_{i,j}=-1.
\end{aligned}$$

We also denote the universal enveloping algebra of $\mathfrak{BD}(n)$ by $BD(n)$.

In [7], it was shown that the affine Yangian associated with $\widehat{\mathfrak{sl}}(n)$ coincides with $A(n)$ if all parameters are zero. If the twisted affine Yangian of types BD exists, we expect that the twisted affine Yangian of types BD should be equal to $BD(n)$ if all the parameters are zero. Moreover, since the twisted affine Yangian should be a coideal of the affine Yangian, it is expected that $BD(n)$ can be embedded into $A(n)$.

By a direct computation, we obtain a homomorphism $\phi\colon BD(n)\to A(n)$ given by

$$B_{i,r}=\begin{cases}E_{i,i+1}v^r-(-1)^rE_{i+1,i}v^r & \text{if } 1\le i\le 2n-1,\\ E_{n,1}uv^r-(-1)^rE_{1,n}u^{-1}v^r & \text{if } i=0,\end{cases}$$

$$H_{i,2r+1}=\begin{cases}2(E_{i,i}-E_{i+1,i+1})v^{2r+1} & \text{if } 1\le i\le 2n-1,\\ 2(E_{n,n}-E_{1,1})v^{2r+1}+2u^{-1}v^{2r+1}du & \text{if } i=0.\end{cases}$$

Theorem 2.2. *The homomorphism ϕ is injective.*

Proof. For (i,j,r,s) such that $E_{i,j}u^rv^s\in A(n)_+$, we define $E^{r,s}_{i,j}\in BD(n)$ inductively as follows:

$$E^{0,s}_{i,i+1}=B_{i,s},E^{1,s}_{n,1}=B_{0,s}$$

and

$$\begin{aligned}
E^{r,s}_{i,j}&=\big[E^{r,s}_{i,j-1},B_{j-1,0}\big]\qquad&&\text{if } j>i+1 \text{ and } 1\le i<j\le n,\\
E^{r,s}_{i,i+1}&=\frac{1}{2}\big[H^r_{i,s},B_{i,0}\big],&&\\
H^r_{i,s}&=\big[E^{r-1,0}_{i,i+1},E^{1,s}_{i+1,i}\big]\qquad&&\text{if } 1\le i\le n-1,\\
H^r_{0,s}&=\big[B_{0,0},E^{r-1,s}_{1,n}\big],&&\\
E^{r,s}_{i,j}&=\big[E^{r,s}_{i,j-1},B_{j-1,0}\big]\qquad&&\text{if } i>j \text{ and } n\ge i\ge j\ge 2,\\
E^{r,s}_{i,1}&=\big[E^{r-1,s}_{i,n},B_{0,0}\big]\qquad&&\text{if } 1\le i\le n-1,\\
E^{r,s}_{n,1}&=\frac{1}{2}\big[H^1_{0,0},E^{r-1,s}_{n,1}\big]\qquad&&\text{for } r\ge 2.
\end{aligned}$$

By a direct computation, we obtain

$$\begin{aligned}
\phi\big(E^{r,s}_{i,j}\big)&=F^{r,s}_{i,j}\qquad\text{if } j>i+1 \text{ and } 1\le i<j\le n,\\
\phi\big(E^{r,s}_{i,i+1}\big)&=F^{r,s}_{i,i+1},
\end{aligned}$$

$$\begin{aligned}
&\phi\big(H^r_{i,s}\big) = F^{r,s}_{i,i} - F^{r,s}_{i+1,i+1} + (r-1)u^{r-1}v^s du - (r-1)(-1)^s u^{-1-r}v^s du \qquad \text{if } 1 \le i \le n-1,\\
&\phi\big(H^r_{0,s}\big) = F^{r,s}_{n,n} - F^{r,s}_{1,1} + u^{r-1}v^s du - (-1)^s u^{-1-r}v^s du \qquad \text{for } r \ge 2,\\
&\phi\big(E^{r,s}_{i,j}\big) = F^{r,s}_{i,j} \qquad \text{if } i > j \text{ and } n \ge i \ge j \ge 2,\\
&\phi\big(E^{r,s}_{i,1}\big) = F^{r,s}_{i,1} \qquad \text{if } 1 \le i \le n-1,\\
&\phi\big(E^{r,s}_{n,1}\big) = F^{r,s}_{n,1} \qquad \text{for } r \ge 2,
\end{aligned}$$

where $F^{r,s}_{i,j} = E_{i,j}u^r v^s - (-1)^s E_{j,i}u^{-r}v^s$. Since $\{\phi(E^{r,s}_{i,j})\} \cup \{\phi(H_{i,2r+1})\}$ are linearly independent, the set $\{E^{r,s}_{i,j}\} \cup \{H_{i,2r+1}\}$ is linearly independent. By the PBW theorem, if we fix an order of the set $\{H_{k,2r+1}, E^{r,s}_{i,j}\}$ satisfying that $H_{k,2r+1} < E^{r,s}_{i,j}$, the set of ordered polynomials derived from $\{H_{k,2r+1}, E^{r,s}_{i,j}\}$ is linearly independent in $BD(n)$. Thus, it is enough to show that these ordered polynomials span $BD(n)$.

Let us take a grading on $BD(n)$ by $\deg(B_{j,r}) = r+1$ and $\deg(H_{j,r}) = r+1$. We denote the graded algebra associated with $BD(n)$ by $\operatorname{gr} BD(n)$ and the images of $\{E^{r,s}_{i,j}\}$ and $\{H^r_{i,s}\}$ in $\operatorname{gr} BD(n)$ by $\{\bar{E}^{r,s}_{i,j}\}$ and $\{\bar{H}^r_{i,s}\}$. The graded algebra $\operatorname{gr} BD(n)$ is generated by $\{\bar{H}_{i,2r+1}, \bar{B}_{i,r}\}$ which satisfy the following relations:

$$\begin{aligned}
\big[\bar{H}_{i,2r+1}, \bar{H}_{j,2s+1}\big] &= 0,\\
\big[\bar{H}_{i,2r+1}, \bar{B}_{j,s}\big] &= 0,\\
\big[\bar{B}_{i,r+1}, \bar{B}_{j,s}\big] &= \big[\bar{B}_{i,r}, \bar{B}_{j,s+1}\big],\\
\big[\bar{B}_{i,r}, \bar{B}_{j,s}\big] &= 0 \qquad \text{if } a^A_{i,j} = 0,\\
\big[\bar{B}_{i,r}, \big[\bar{B}_{i,s}, \bar{B}_{j,u}\big]\big] &= 0 \qquad \text{if } a^A_{i,j} = -1.
\end{aligned}$$

Let L_+ be the subalgebra generated by $\{\bar{B}_{i,r}\}$. We can decompose $\operatorname{gr} D(n)$ into the tensor product of $\mathbb{C}[H_{i,2r+1}]$ and L_+. By Theorem 1.3, we have a surjective homomorphism from $A(n)_+$ to L_+. By definition, $\bar{E}^{r,s}_{i,j}$ corresponds to $E_{i,j}u^r v^s$, $H^r_{i,s}$ corresponds to $(E_{i,i} - E_{i+1,i+1})u^r v^s + (r-1)u^r v^s du$ and $H^r_{0,s}$ corresponds to $(E_{n,n} - E_{1,1})u^r v^s + u^{r-1}v^s du$. Since the set of these elements is a basis of $\bigoplus_{s\ge 0}\bigoplus_{\alpha\in\Delta^+}\sum_{k=1}^{\dim \mathfrak{g}_\alpha} x^k_\alpha v^s \oplus \Omega_+$, we find that the set of ordered polynomials derived from $\{H_{k,2r+1}, E^{r,s}_{i,j}\}$ spans $BD(n)$. Thus, ϕ becomes injective. □

References

[1] Belliard S., Regelskis V., Drinfeld J presentation of twisted Yangians, *SIGMA* **13** (2017), 011, 35 pages, arXiv:1401.2143.

[2] Brown J., Twisted Yangians and finite W-algebras, *Transform. Groups* **14** (2009), 87–114, arXiv:0710.2918.

[3] Brundan J., Kleshchev A., Shifted Yangians and finite W-algebras, *Adv. Math.* **200** (2006), 136–195, arXiv:math/0407012.

[4] De Sole A., Kac V.G., Valeri D., A Lax type operator for quantum finite W-algebras, *Selecta Math. (N.S.)* **24** (2018), 4617–4657, arXiv:1707.03669.

[5] Drinfeld V.G., Hopf algebras and the quantum Yang–Baxter equation, *Soviet Math. Dokl.* **32** (1985), 256–258.

[6] Drinfeld V.G., A new realization of Yangians and of quantum affine algebras, *Soviet Math. Dokl.* **36** (1988), 212–216.

[7] Guay N., Affine Yangians and deformed double current algebras in type A, *Adv. Math.* **211** (2007), 436–484.

[8] Guay N., Nakajima H., Wendlandt C., Coproduct for Yangians of affine Kac–Moody algebras, *Adv. Math.* **338** (2018), 865–911, arXiv:1701.05288.

[9] Harako S., Ueda M., A new presentation of the twisted Yangian of type D, arXiv:2507.00350.

[10] Kac V.G., Infinite-dimensional Lie algebras, 3rd ed., Cambridge University Press, Cambridge, 1990.

[11] Kassel C., Kähler differentials and coverings of complex simple Lie algebras extended over a commutative algebra, *J. Pure Appl. Algebra* **34** (1984), 265–275.

[12] Lu K., Minimalistic presentation and coideal structure of twisted Yangians, *Comm. Math. Phys.* **407** (2026), 99, 38 pages, arXiv:2511.07136.

[13] Lu K., Peng Y.N., Tappeiner L., Topley L., Wang W., Shifted twisted Yangians and finite W-algebras of classical type, arXiv:2505.03316.
[14] Lu K., Wang W., Zhang W., A Drinfeld type presentation of twisted Yangians, *Represent. Theory* **29** (2025), 838–870, arXiv:2308.12254.
[15] Lu K., Wang W., Zhang W., Affine $\imath$quantum groups and twisted Yangians in Drinfeld presentations, *Comm. Math. Phys.* **406** (2025), 98, 36 pages, arXiv:2406.05067.
[16] Lu K., Zhang W., A Drinfeld type presentation of twisted Yangians of quasi-split type, *Commun. Contemp. Math.*, to appear, arXiv:2408.06981.
[17] Matsuo A., Nagatomo K., Tsuchiya A., Quasi-finite algebras graded by Hamiltonian and vertex operator algebras, in Moonshine: the First Quarter Century and Beyond, *London Math. Soc. Lecture Note Ser.*, Vol. 372, Cambridge University Press, Cambridge, 2010, 282–329, arXiv:math/0505071.
[18] Moody R.V., Rao S.E., Yokonuma T., Toroidal Lie algebras and vertex representations, *Geom. Dedicata* **35** (1990), 283–307.
[19] Olshanskii G.I., Twisted Yangians and infinite-dimensional classical Lie algebras, in Quantum Groups (Leningrad, 1990), *Lecture Notes in Math.*, Vol. 1510, Springer, Berlin, 1992, 104–119.
[20] Ueda M., Twisted affine Yangian and rectangular W-algebra of type D, *Algebr. Represent. Theory* **28** (2025), 1195–1224, arXiv:2107.09999.
[21] Ueda M., Suggestion of a definition of the twisted affine Yangian of type D, arXiv:2512.06340.